

**Blue Origin Mark-1 Payloads For Lunar Surface Science.** E. B. Wagner<sup>1</sup>, S.B. Martinez<sup>1</sup>, and J.A. Cortese<sup>1</sup>, <sup>1</sup>Blue Origin (21218 76<sup>th</sup> Ave S, Kent, WA 98032; [lunarsales@blueorigin.com](mailto:lunarsales@blueorigin.com)).

**Introduction:** Blue Origin is committed to building a road to space so that the next generation can access the limitless resources of space and improve life here on Earth. With Blue Moon, the team has defined and developed a state-of-the-art lunar lander architecture to provide reliable, low-cost launch and commercial lunar transport vehicles that will enable a thriving cislunar and lunar surface ecosystem. Blue Moon Mark-1 (MK1) is Blue Origin's single-launch, expendable cargo lander, designed to deliver payloads and multiple metric tons into lunar orbit and down to the surface of the Moon, providing safe, reliable, and affordable access to the lunar environment. This sizable capacity is paired with significant power, thermal, and data services, and precision landing capabilities to any lunar destination, making MK1 a market-leading platform for science, technology, exploration, and commerce.

The MK1 Payload User's Guide (PUG) [1] provides a technical point of entry for users of both hosted and deployed payloads, describing MK1's performance, environments, requirements, interfaces, facilities, and operations. Researchers can engage with interests in sustainable lunar presence, novel scientific inquiries, demonstrating space technologies, commercializing services and products on the Moon, or engaging communities back on Earth.

**Technical Capabilities:** MK1 leverages the seven-meter fairing of the New Glenn launch vehicle to provide large lunar cargo transport, including industrial-scale infrastructure payloads. MK1 is designed to deliver up to three metric tons to anywhere on the lunar surface. The first Pathfinder Mission in 2025 will be a demonstration mission to prove out the design of the MK1 system. Follow-on flights will be available to payload customers to meet research needs in biological and physical sciences, astrophysics, heliophysics, and technology maturation.

Standard payload interfaces are available on both the aft module and top deck. Each interface is served with structural mounts, regulated DC power, RS-485/RS-422 data handling, and direct to Earth communications. Additional optional services include active thermal management, offloading to the lunar surface, custom interfaces, and mission-unique design and analysis support.

This talk will present updated plans and technology development efforts currently underway for MK1, describe how these efforts will serve the science and technology development community, and provide high-level descriptions of how the research community can manifest payloads for lunar access.

**References:**

[1] Blue Origin (2023) *MLMA-96737, Blue Moon MK1 Payload User's Guide*.

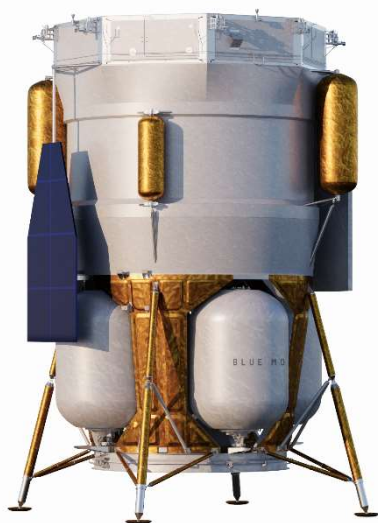


Figure 1: Blue Origin's Mark-1 Lunar Lander supports up to three metric tons of lunar surface science payload